Risk and Return Revisit

Jinfan Zhang

Chinese University of Hong Kong (Shenzhen)

Definition of return

- Return=(what you get tomorrow)/what you paid today
  or, R=(value at time t+1)/(value at time t)

$$R_{t+1} = \frac{P_{t+1} + CF_{t+1}}{P_t}$$

clean/dirty price

eg. (Bond market)

Jan.1 12%

Jul.1 101 101+6=107

Oct.1 102 102+9=111

- What you get tomorrow is the price of the asset plus any cash flow it pays off
  - Ex: for a stock CF=dividend, for a bond CF=coupon payment
- This is often called **gross return**

Gross vs. net return

- We will typically work with **net returns**, r

$$r_{t+1} = \frac{P_{t+1} + CF_{t+1}}{P_t} - 1$$

- Example, suppose you buy a stock at P=$100, tomorrow's price is $101 and it pays a $3 dividend
  - Gross return = (101+3)/100=1.04
  - Net return =1.04-1=0.04 or 4%
  - 4%, the net return, is often a more convenient number to think about than 1.04

Expected Return

expected return = looking forward
realized return = looking backward

- Actual returns are not known ahead of time, so we will often compute **expected returns**

$$E\left[r_{t+1}\right] = \frac{E\left[P_{t+1}\right] + E\left[CF_{t+1}\right]}{P_t} - 1$$

- This tells us what the return is likely to be *on average*
- **Realized returns** (what you actually get) are often very different from expected returns
  - Example: stock price today is $100, equal chance it will go up to either $150 or down to $50. Realized return will be either +50% or -50%. Expected return is E[r]=0.5*(50%)+0.5*(-50%)=0.

Multi-period returns and compounding

- $R_{t,t+1}$ gross return from time t to t+1
- $R_{t+1,t+2}$ gross return from time t+1 to t+2
- $R_{t,t+2}$ gross return from time t to t+2
- $r_{t,t+2} = R_{t,t+2} - 1$ net return from t to t+2

Multi-period returns and compounding

- Then $$R_{t,t+2} = R_{t,t+1}R_{t+1,t+2}$$

$$(1+r_{t,t+2}) = (1+r_{t,t+1})(1+r_{t+1,t+2})$$

$= 1 + r_{t,t+1} + r_{t+1,t+2} + r_{t,t+1}\, r_{t+1,t+2}$

$\approx 1 + r_{t,t+1} + r_{t+1,t+2}$

- This is the standard compounding formula $\Rightarrow r_{t,t+2} = r_{t,t+1} + r_{t+1,t+2}$
  - Example: Suppose you earn **10%** in year 1 and **20%** in year 2.
  - Your 2-year net return is (1+0.1)(1+0.2)-1=**32%**

Multi-period returns and compounding

- For a stock

2 cash flow

$$R_{t,t+2} = \frac{P_{t+2} + D_{t+2} + D_{t+1}\dfrac{P_{t+2} + D_{t+2}}{P_{t+1}}}{P_t}$$

dividend be reinvested (immediately)

$$R_{t,t+2} = \frac{P_{t+2} + D_{t+2}}{P_{t+1}} \frac{P_{t+1} + D_{t+1}}{P_t} = R_{t,t+1} R_{t+1,t+2}$$

Real vs. nominal returns

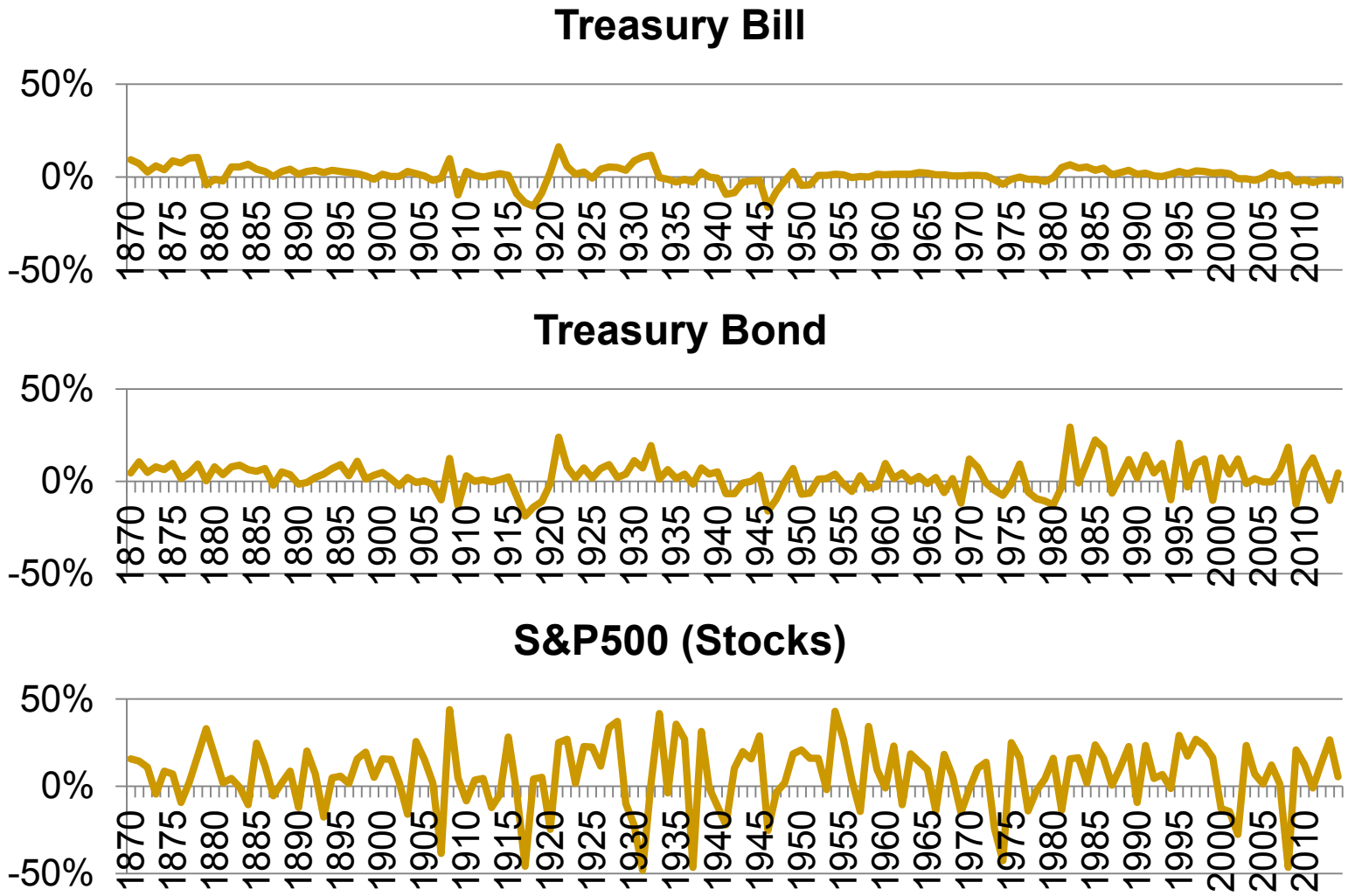
- So far we have been working with nominal returns
- To convert a nominal return to a real return we divide by the rate of inflation, π

$$(1+r_{real}) = \frac{(1+r_{nom})}{(1+\pi)} = \frac{(1+r_{nom})(1-\pi)}{(1+\pi)(1-\pi)} = \frac{1+r_{nom}-\pi-r_{nom}\cdot\pi}{1-\pi^2}$$

$$\Rightarrow 1+r_{nom} \approx \frac{1+r_{nom}-\pi}{1}$$

- Approximation $r_{real} \approx r_{nom} - \pi$

Real Returns Historically



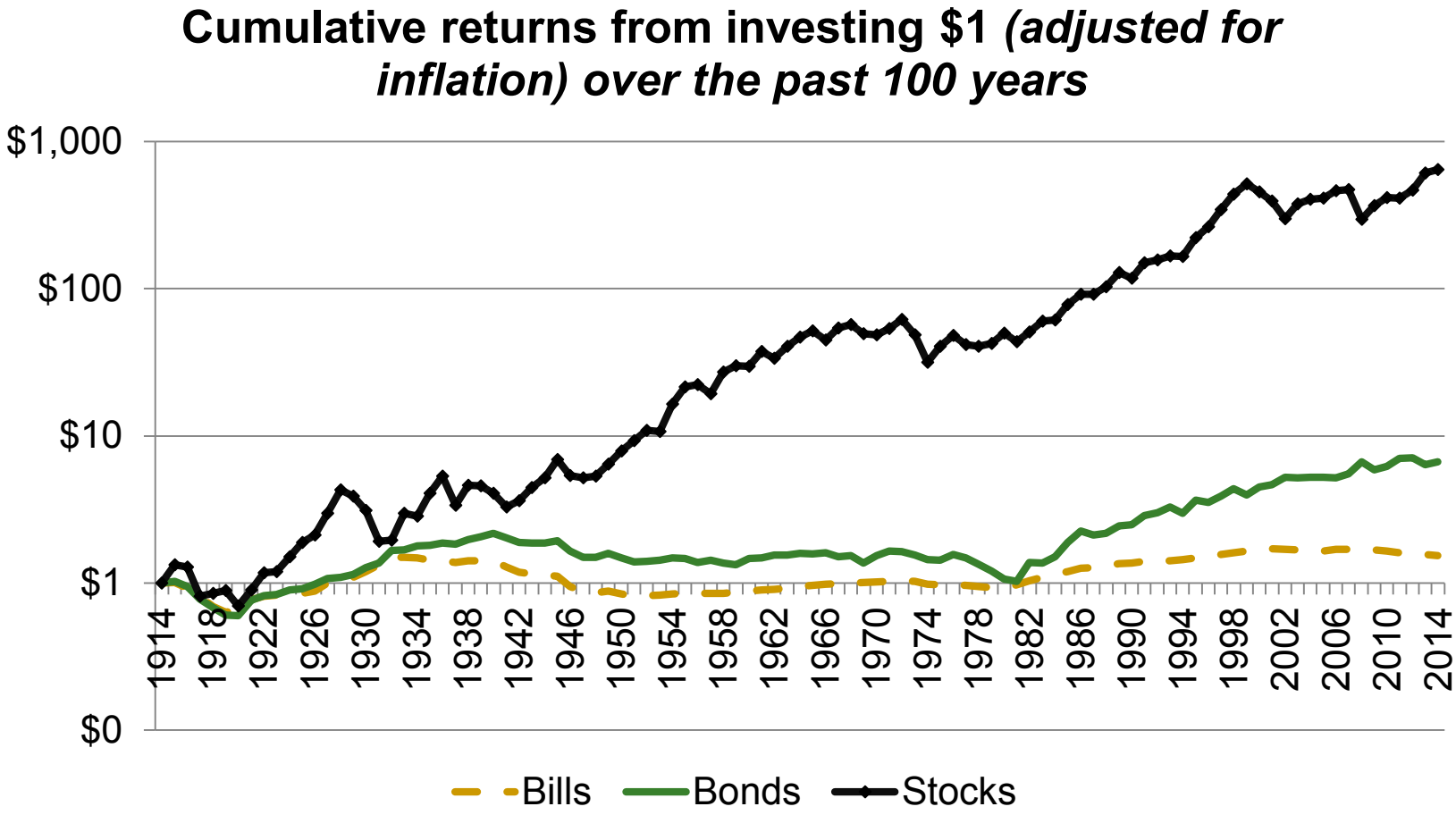
Historical Returns

- What have the real, annual, realized returns on securities been historically?

Period	T-Bills	T-Bonds	Stocks (S&P500)
1870-1925	2.6%	2.8%	6.5%
1926-2014	0.6%	2.2%	6.6%

- **Equity premium** is the average return of stocks minus the average return on T-Bills: It is around 6% historically.

Equity premium



Asset Allocation Question

- Does this mean we should put all our money into stocks?
- Why? Why not?
- How should we think about this question?

Risk and Return

- Risk is uncertainty about the future
- Over the long run stocks do better on average, but year to year their annual returns are more volatile or uncertain

	T-Bills	T-Bonds	Stocks (S&P500)
Average return	0.6%	2.2%	6.6%
Volatility	3.0%	6.1%	19.1%
Worst year	-15.2%	-16.0%	-48.0%

- Stocks are “riskier”
- Example: the S&P500 dropped over 40% in 2008

Risk and Return

- Human beings are *mostly* **risk averse**, i.e. dislike risk
- To be willing to hold a riskier security, they must receive a higher average return as a reward for doing so
  - This is why stocks have higher average returns than bonds
  - Ex ante vs. ex post
- "Average return is compensation for risk" is a good way of thinking about long-term patterns in average returns
- We will discuss how to measure risk more precisely later in the course

Expected Return

- Expected returns

$$E(r) = \sum_{s} p(s)r(s)$$

  - p(s) = Probability of a state
  - r(s) = Return if a state occurs
  - s = State

Scenario Returns: Example

State	Prob. of State	*r* in State
Excellent	.25	0.3100
Good	.45	0.1400
Poor	.25	-0.0675
Crash	.05	-0.5200

$E(r) = (.25)(.31) + (.45)(.14) + (.25)(-.0675) + (0.05)(-0.52)$

$= .0976$ or 9.76%

Standard Deviation

- Variance (VAR):

$$\sigma^2 = \sum_s p(s)[r(s) - E(r)]^2$$

- Standard Deviation (STD):

$$\text{STD} = \sqrt{\sigma^2}$$

VAR and STD: Example

- VAR calculation:

$$\sigma^2 = .25(.31 - 0.0976)^2 + .45(.14 - .0976)^2 + .25(-0.0675 - 0.0976)^2 + .05(-.52 - .0976)^2$$
$$= .038$$

- STD calculation:

$$\sigma = \sqrt{.038}$$
$$= .1949$$

Time Series Analysis of Past Rates of Return

- You can only observe a time series of past returns
  - You cannot see all possible states
- True means and variances are unobservable because we don't actually know possible scenarios like the one in the examples
- So we must estimate the means and variances

Arithmetic vs. Geometric Average

- Arithmetic Average

$$E(r) = \frac{1}{n}\sum_{i=1}^{n} r_i$$

- Arithmetic Average is an unbiased estimate of the **expected future return**

- What is the **actual** performance of the **past** sample?

- Geometric (Time-Weighted) Average

$$g = \left((1+r_1)(1+r_2)\ldots(1+r_n)\right)^{1/n} - 1$$

Arithmetic vs. Geometric Average

- Your returns in the last 2 years are 10%, -10%.
- What is your expected return next year?
  - **Arithmetic Average: 0%**
- What was your return over the last two years?
  - $1*(1+10%)*(1-10%)=0.99, that is -1% over 2 years,
  - Or, SQRT(0.99)-1=-0.5% per year
  - **The geometric average return is -0.5%**
- Asymmetry: negative return has a larger effect
  - You had a -50% return last year, what is the return you need this year to break even?

Arithmetic vs. Geometric Average

- What determines the difference between the two?
- Case 1: returns are not volatile: 9%, 11%
  - Arithmetic average: 10%
  - Geometric average: 9.995%
- Case 2: returns are volatile: -90%, 110%
  - Arithmetic average: 10%
  - Geometric average: -54%
- The difference is larger when returns are more volatile

Estimating Variance and Standard Deviation

- Estimated Variance

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\left[r_i - \bar{r}\right]^2$$

- Unbiased estimated standard deviation

$$\hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left[r_i - \bar{r}\right]^2}$$

Sharpe Ratio

- Excess Return
  - The difference in any particular period between the **actual** rate of return on a risky asset and the actual risk-free rate
- Risk Premium
  - The difference between the **expected** return on a risky asset and the risk-free rate
- Sharpe Ratio

$$\frac{\text{Risk premium}}{\text{SD of excess returns}}$$

1. CAPM

$$E(\tilde{r}_i) = r_f + \beta_i\left(E(\tilde{r}_M) - r_f\right)$$

$$\beta_i = \frac{cov(\tilde{r}_i, \tilde{r}_M)}{Var(\tilde{r}_M)}$$

$$E(\tilde{r}_p) = r_f + \beta_p\left(E(\tilde{r}_M) - r_f\right)$$

$$E(\tilde{r}_p) - r_f = \beta_p\left(E(\tilde{r}_M) - r_f\right)$$

$$SD(\tilde{r}_p - r_f) = SD(\tilde{r}_p)$$

2. APT (one factor model)

$$\tilde{r}_i = \alpha_i + \beta_i \cdot \tilde{F}_1 + \tilde{\varepsilon}_i$$

$$\tilde{r}_p = \alpha_p + \beta_p \cdot \tilde{F}_1$$

$$SD(\tilde{r}_p - r_f) = SD(\tilde{r}_p) = \beta_p \cdot \sigma$$

$$\sigma = SD(\tilde{F}_1)$$

According to APT:

$$\alpha_p = r_f + \beta_p \cdot \lambda$$

$$SR = \frac{\alpha_p - r_f}{\beta_p \cdot \sigma} = \frac{\beta_p \cdot \lambda}{\beta_p \cdot \sigma} \rightarrow \text{a constant}$$

$$\tilde{F}_1 = \tilde{r}_M - E(r_M)$$

3. APT (multiple/2 factor model)

$$\tilde{r}_i = \alpha_i + \beta_1 \tilde{F}_1 + \beta_2 \tilde{F}_2 + \varepsilon_i$$

$$\tilde{r}_p = r_f + \beta_1 \lambda_1 + \beta_2 \lambda_2$$

$$Var(\tilde{r}_i) = \beta_1^2 \sigma_{F_1}^2 + \beta_2^2 \sigma_{F_2}^2 + 2\beta_1\beta_2 Cov(\tilde{F}_1, \tilde{F}_2)$$

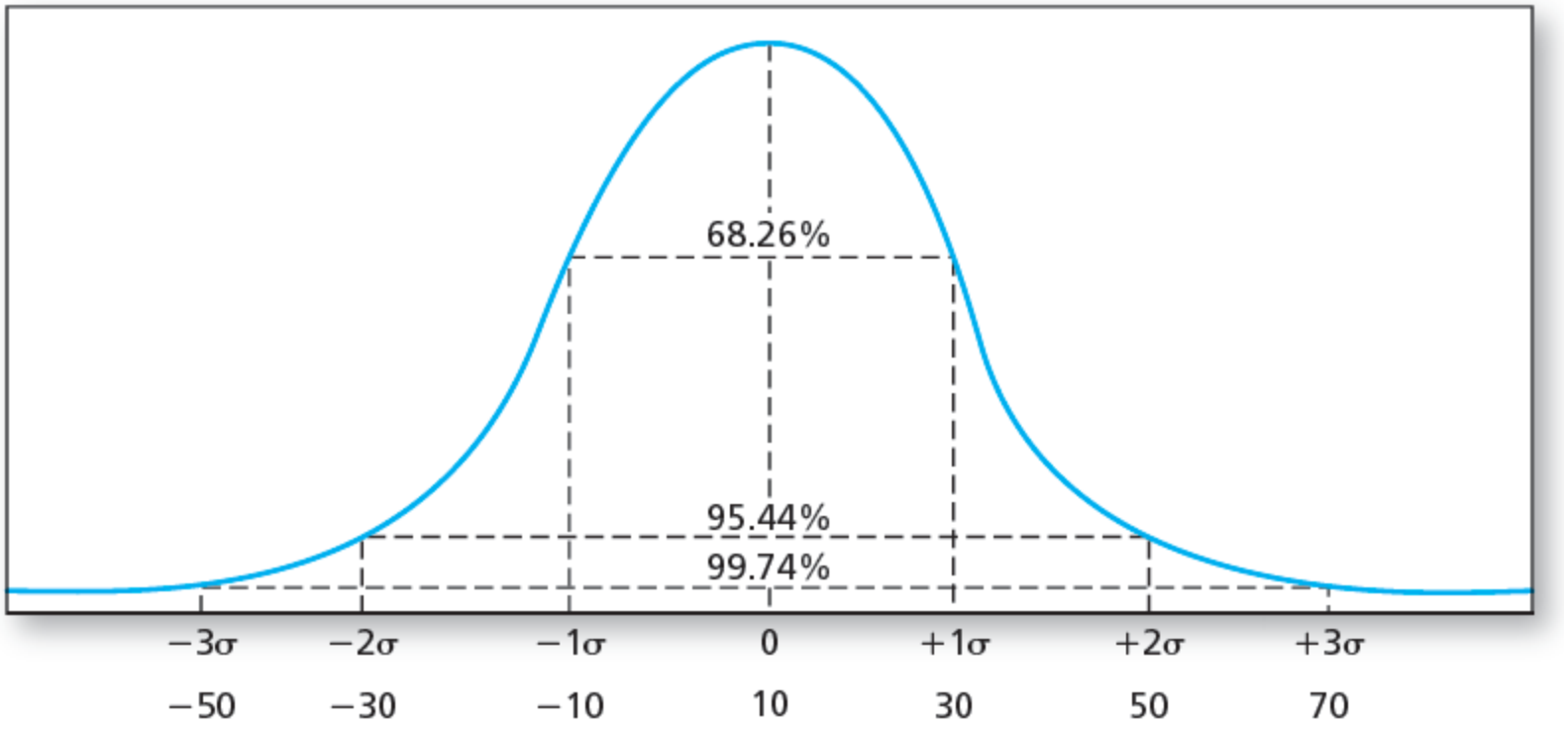
$$SD = \sqrt{Var(r_i)}$$

$$SR = \frac{\beta_1\lambda_1 + \beta_2\lambda_2}{\sqrt{\beta_1^2 \sigma_{F_1}^2 + \beta_2^2 \sigma_{F_2}^2 + 2\beta_1\beta_2 Cov(\tilde{F}_1, \tilde{F}_2)}}$$

The Normal Distribution

- **If returns are normally distributed**
  - Standard deviation is a good measure of risk
  - Portfolio returns are also normal
  - Future scenarios can be estimated using only the mean and the standard deviation
  - The dependence of returns across securities can be summarized using only the pairwise correlation coefficients

The PDF of a Normal Distribution



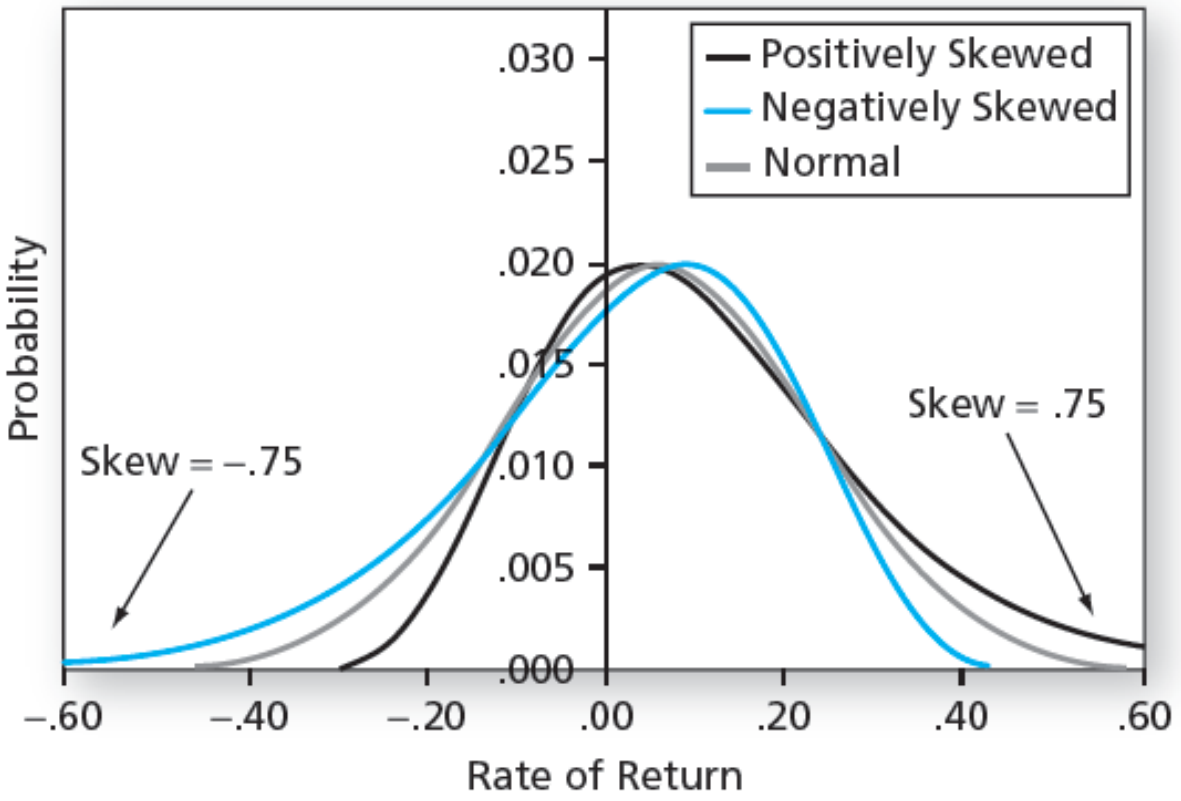
Mean = 10%, SD = 20%

Normality and Risk Measures

- What if excess returns are not normally distributed?
  - Standard deviation is no longer a complete measure of risk
  - Sharpe ratio is not a complete measure of portfolio performance
  - Need to consider skewness and kurtosis

Normal and Skewed Distributions

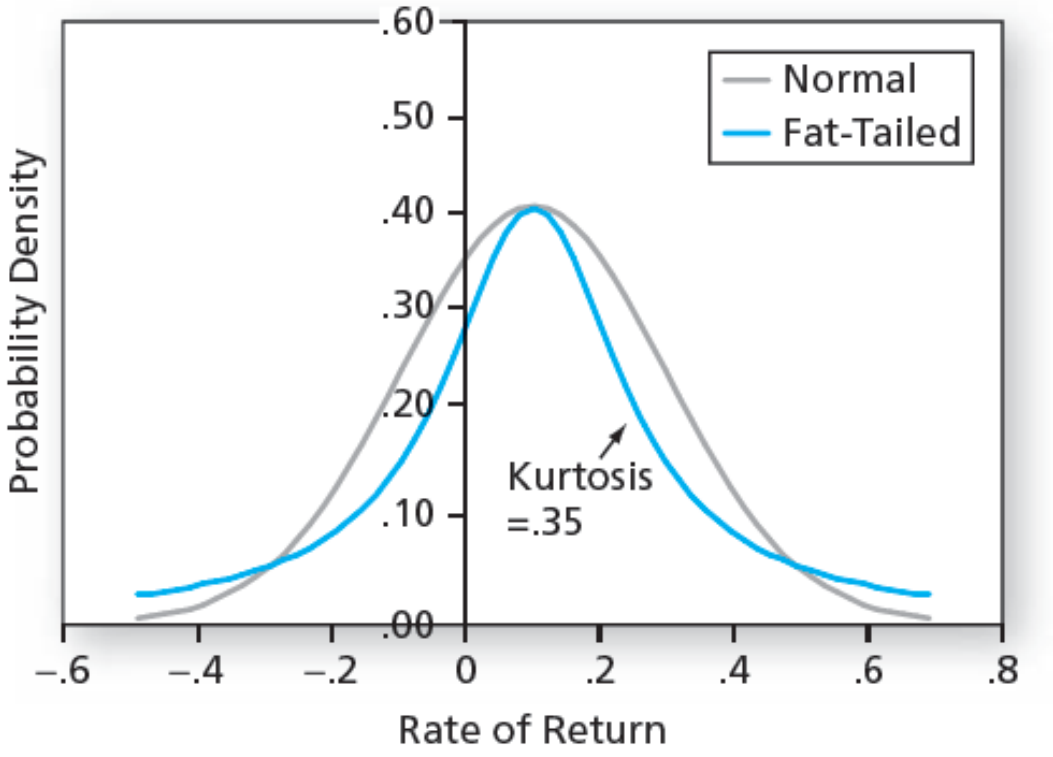
eg. $\tilde{r}_i = \alpha_i + \beta_{i1}(\tilde{r}_M - \bar{r}_M) + \beta_{i2}\tilde{F}_2$



Mean = 6%, SD = 17%

$$g_1 = \frac{\sum_{i=1}^{N}(Y_i - \bar{Y})^3/N}{s^3}$$

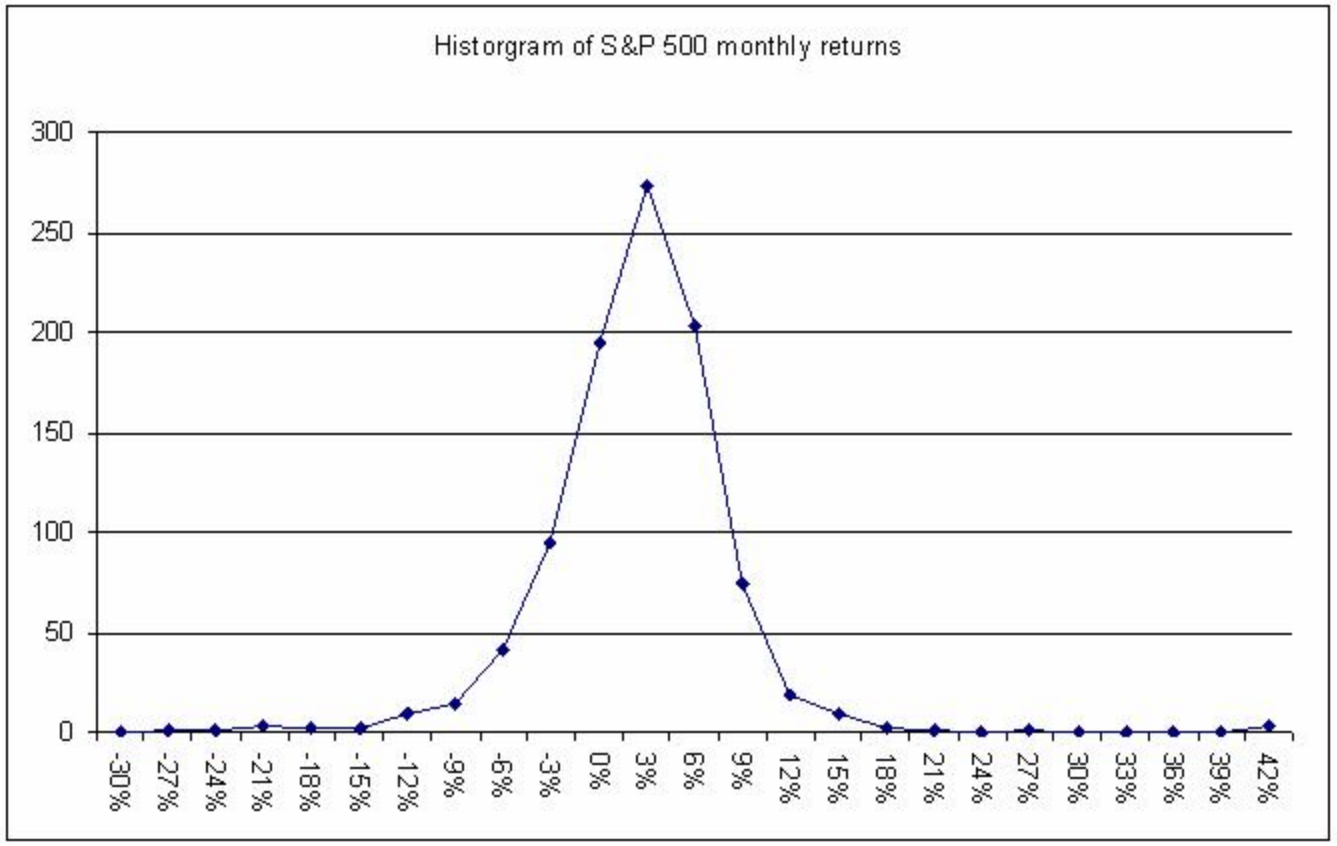
Normal and Fat-Tailed Distributions



Mean = .1, SD = .2

$$\text{kurtosis} = \frac{\sum_{i=1}^{N}(Y_i - \bar{Y})^4/N}{s^4} - 3$$

Example: S&P 500 (1926-2004)



Normality and Risk Measures

- Value at Risk (VaR)
  - Loss corresponding to a very low percentile of the entire return distribution, such as the fifth or first percentile return
  - Example: a 5% VaR of $1M: The chance of losing **more than** or equal to $1M is 5%.

- Expected Shortfall (ES)
  - Also called conditional tail expectation (CTE), focuses on the expected loss in the worst-case scenario (left tail of the distribution)
  - More conservative measure of downside risk than VaR

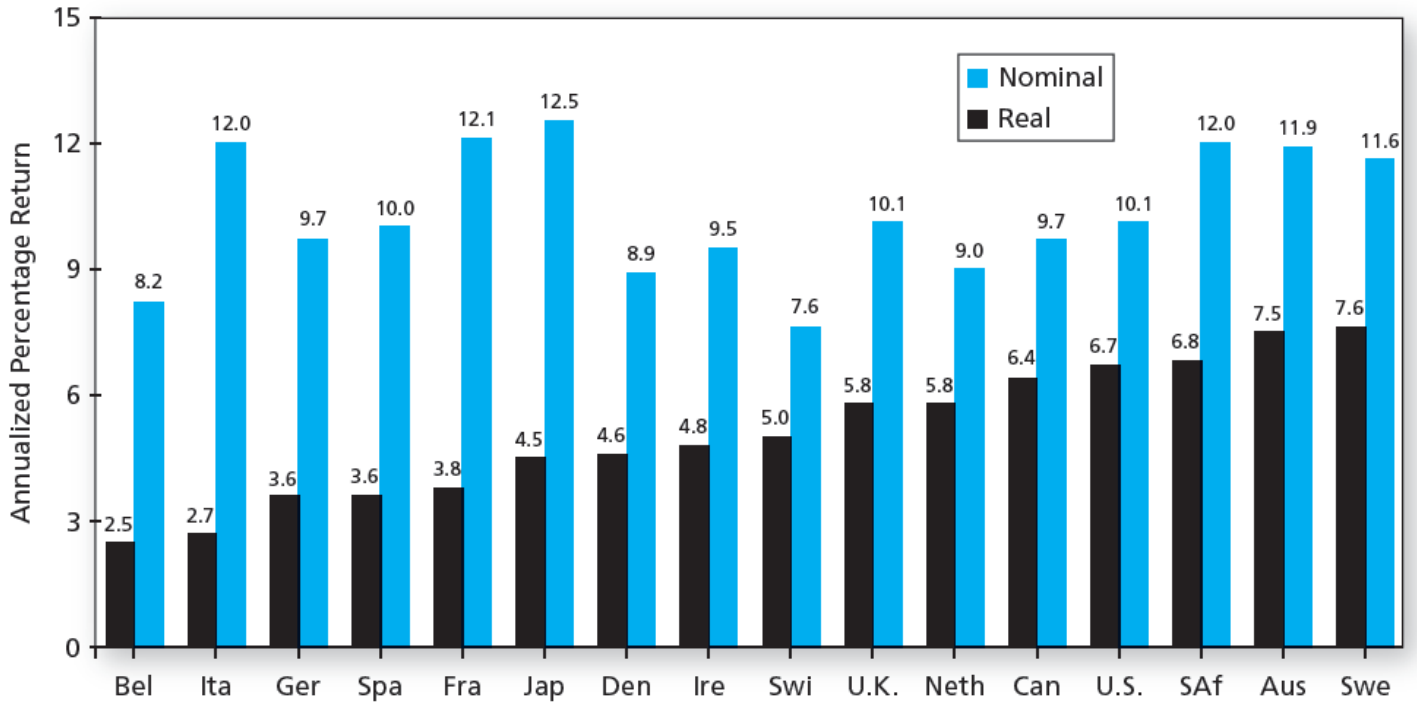
Normality and Risk Measures

- Lower Partial Standard Deviation (LPSD)
  - Similar to usual standard deviation, but uses only negative deviations from the risk-free return, thus, addressing the asymmetry in returns
- Sortino Ratio (replaces Sharpe Ratio)
  - The ratio of average excess returns to LPSD

Historic Returns on Risky Portfolios

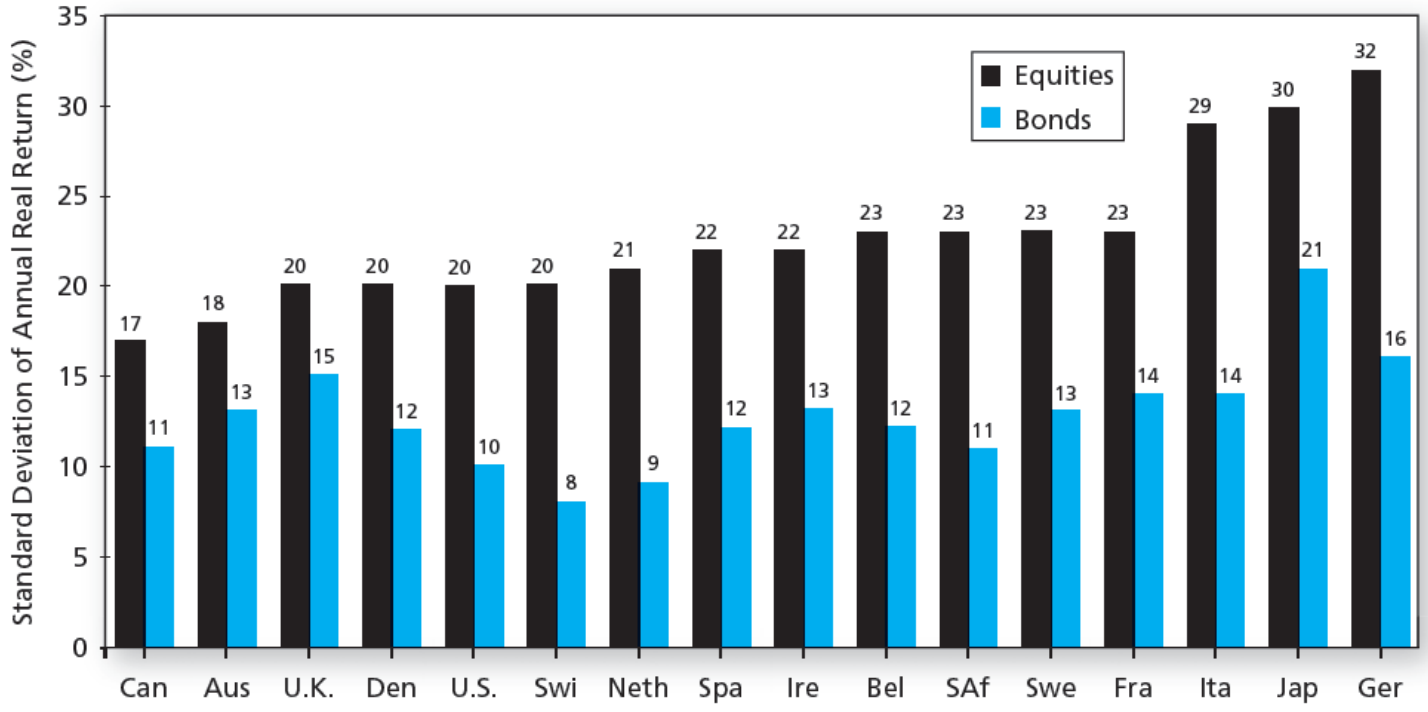
- The second half of the 20th century, politically and economically the most stable sub-period, offered the highest average returns
- Firm capitalization is highly skewed to the right: Many small but a few gigantic firms
- Average realized returns have generally been higher for stocks of small rather than large capitalization firms

Nominal and Real Equity Returns Around the World, 1900-2000



Source: Dimson, March and Staunton (2001), The Triumph of the Optimists: 101 Years of Global Investment Returns

SD of Real Equity & Bond Returns Around the World, 1900-2000



Source: Dimson, March and Staunton (2002), The Triumph of the Optimists: 101 Years of Global Investment Returns

Price: The Present Value Formula

- When investing in an asset, we have in mind an average rate of return that is fair for that asset
  - The "expected return" for an asset $r = E[r_{t+1}]$
  - r= riskless interest rate + risk premium
  - = real interest rate + expected inflation + risk premium
- r, the fair average return, is also called:
  - the "expected return"
  - the "discount rate"
  - the "required return"

The Present Value Formula

- We can rearrange the definition of return to obtain an important pricing formula, the present value formula

$$1+r_{t+1}=\frac{CF_{t+1}+P_{t+1}}{P_t}$$

$$1+r=\frac{E\left[CF_{t+1}\right]+E\left[P_{t+1}\right]}{P_t}$$

$$P_t=\frac{E\left[CF_{t+1}\right]+E\left[P_{t+1}\right]}{1+r}$$

$$P_{t+1}=\frac{E\left[CF_{t+2}\right]+E\left[P_{t+2}\right]}{1+r}$$

The Present Value Formula

- Substituting, we get

$$P_t = \frac{E[CF_{t+1}]}{1+r} + \frac{E[CF_{t+2}]}{(1+r)^2} + \frac{E[P_{t+2}]}{(1+r)^2}$$

- and so on, until

$$P_t = \frac{E[CF_{t+1}]}{1+r} + \frac{E[CF_{t+2}]}{(1+r)^2} + \frac{E[CF_{t+3}]}{(1+r)^3} + \ldots$$

The Present Value Formula

$$P_t = \frac{E[CF_{t+1}]}{1+r} + \frac{E[CF_{t+2}]}{(1+r)^2} + \frac{E[CF_{t+3}]}{(1+r)^3} + \ldots$$

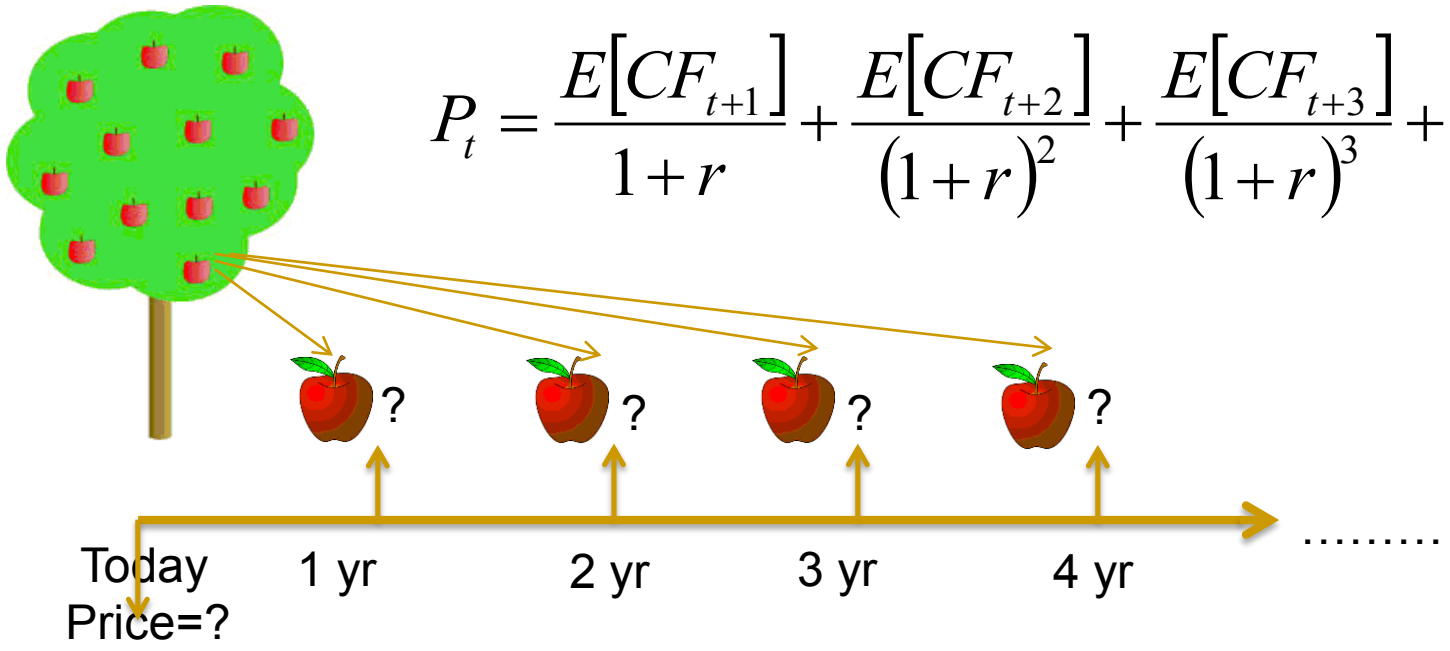
- This formula says that the price, or value, of an asset is the sum of its discounted expected future cash flows
  - This is why the expected return r is often called a "discount rate"

- This formula is incredibly useful in finance for a range of applications (i.e., valuation, capital budgeting, etc.)

The Present Value Formula

- Think of a stock like a tree which yields some uncertain amount of fruit each year ("dividends")

$$P_t = \frac{E[CF_{t+1}]}{1+r} + \frac{E[CF_{t+2}]}{(1+r)^2} + \frac{E[CF_{t+3}]}{(1+r)^3} + \ldots$$



The Present Value Formula

- What is the price, or value, of the tree?
- It is the present value of all the fruit we expect to get
- We “discount” future fruit because it is risky / uncertain and because we are impatient

The Present Value Formula

- Special case: cash flows are expected to grow at a constant rate g

$$P_t = \frac{E[CF_{t+1}]}{1+r} + \frac{E[CF_{t+2}]}{(1+r)^2} + \frac{E[CF_{t+3}]}{(1+r)^3} + \cdots$$

$$P_t = \frac{CF_t(1+g)}{r-g}$$

$$P_t = \frac{CF_{t+1}}{r-g}$$

Understanding changes in asset prices

- When the price of an asset moves, it can be due to:
  - Changes in expectations about future cash flows $E[CF_{t+k}]$
  - Changes in the discount rate r
    - Changes in the real interest rate
    - Changes in the rate of inflation
    - Changes in the risk premium
  - Changes in the risk premium can be due to:
    - Changes in the amount of risk
    - Changes in the price of risk

$$P_t = \frac{E(CF_t)_{+1}}{r - g}$$

Understanding the 1995-2000 surge in stock prices

- From 1995-2000 stock prices nearly tripled
- How can we understand this?
  - An increase in expected future cash flows
  - A fall in the discount rate due to
    - A fall in the risk premium
      - People decided that stocks were less risky
      - People became less risk averse
    - A fall in the real interest rate
    - A fall in inflation

$$P_t = \frac{E(CF_t)_{+1}}{r - g}$$

- Note: the fall in real interest rate and inflation were likely small effects

Valuing the stock market with the PV formula

- Given a reasonable risk premium, what is a fair value for the stock market?

$$\frac{P_t}{E_t} \approx \frac{1}{r - g} = \frac{1}{r_f + RP - g}$$

- Given todays prices, what is the likely average risk premium going to be, going forward

$$RP \approx g + \frac{E_t}{P_t} - r_f$$

- Historically, a high D/P or E/P ratio implies significant high stock returns

Understanding economic reports

- Inflation news
- Bonds
  - If inflation is higher than expected, bonds fall
  - The discount rate goes up
  - Lenders raise interest rates to cover the rise in inflation
- How about stocks?

Understanding economic reports

- Stocks usually do not react as much to inflation news
- Inflation higher than expected
  - The discount rate goes up
  - Expected cash flows increase (in nominal terms)
  - Effects tend to cancel each other out

Understanding economic reports

- Employment reports and bonds
- If unemployment figures are better than expected (i.e., good news), bonds fall
- The discount rate goes up
- Firms borrow more in a stronger economy, pushing the riskless rate up

Bond Prices Post Steep Losses on Strong Economic News

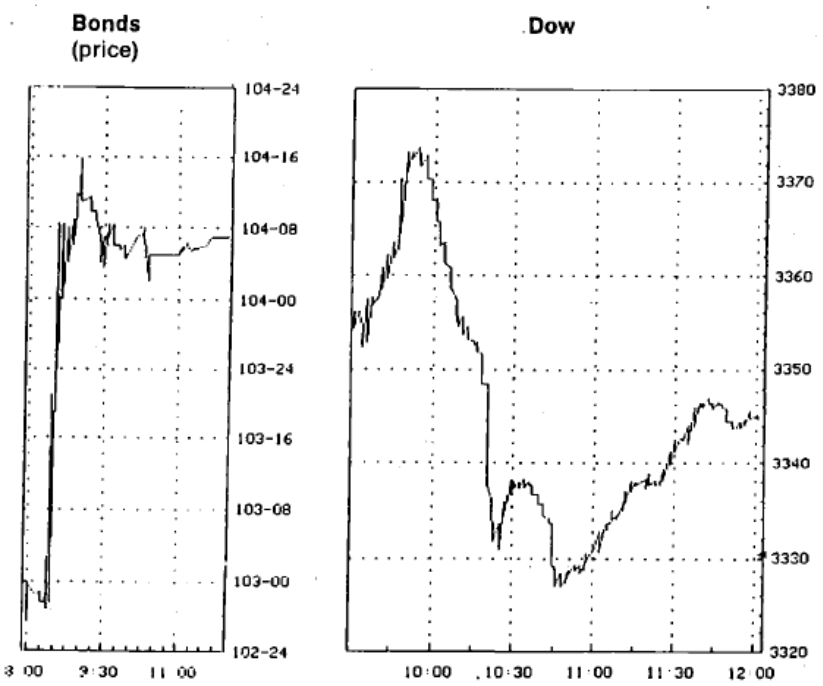
Understanding economic reports

- Employment reports and stocks
- Stocks *typically* do not react as much to employment reports
- If unemployment figures are better than expected (i.e., good news)
  - Again, the discount rate will rise
  - But the stronger economy means that cash flows will rise as well
  - The two effects tend to cancel out (on average)

Understanding economic reports

- Reaction of stocks / bonds to *bad* employment news

FIGURE 13–1
Weak Employment Report and Fed Ease



Summary

- Gross return
- Net return
- Expected return
- Realized return
- Arithmetic average
- Geometric average
- Variance
- Standard deviation
- Equity premium, Sharpe ratio
- VaR, ES, LPSD, Sortino ratio

Summary

- Present value formula:

$$P_t = \frac{E[CF_{t+1}]}{1+r} + \frac{E[CF_{t+2}]}{(1+r)^2} + \frac{E[CF_{t+3}]}{(1+r)^3} + \ldots$$

  - When cash flows grow at rate g: $P_t \approx \frac{CF_t}{r-g}$

- It helps us understand movements in asset prices

- Reference: Chapter 5, BKM